The Combination Between Deep Learning and Ensemble Stacking for A Fast and Accurate Detection of Diabetic Retinopathy Using Fundus Images

Anass TIBARI
RITM Laboratory, CED Engineering Sciences Hassan II University
Casablanca, Morocco
anas.tibari@gmail.com

Mohammed RIDOUANI
RITM Laboratory, CED Engineering Sciences Hassan II University
Casablanca, Morocco
mohammed.ridouani@gmail.com

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***Abstract*— Diabetic retinopathy (DR) is the leading cause of human vision loss in the world. To slow the progression of the disease we need early detection and diagnosis. Hence, rapid detection and accurate classification of DR is crucial in patient care. The use of Deep learning (DL) alone turns out to be slow and expensive, also the use of a single Machine Learning (ML) algorithm for classification leaves a significant margin of error. The detection of the disease also proves to be more difficult, especially in its early stages, this is due to the characteristics of DR less clear and visible on images. Therefore, the combination of Deep Learning and ensemble stacking was a relevant solution that we chose to enhance the ability to detect the disease in its early stages. In our research, we introduce an innovative method utilizing an automated model to identify and classify the existence and stage of severity of DR directly from fundus images. We utilized both of Machine Learning algorithms and a Deep Learning model; to achieve an accurate and fast detection we used the transfer learning and ensemble stacking as techniques. The proposed model uses a dataset published on Kaggle; our solution achieves an accuracy of 99.50%.**

Keywords— Diabetic retinopathy, fundus images, VGG-16, transfer learning, ensemble stacking, machine learning, deep learning, classification, diagnosis.

I. Introduction

Diabetic Retinopathy is a major side effect of diabetes that affects the retinal blood vessels and can result in vision loss if left untreated. Early detection of DR plays a crucial role in preventing further progression of the disease and preserving vision [1]. Regular screening and timely diagnosis of DR are essential for diabetes patients to avoid the risk of severe complications of DR also early detection is vital in the early stages of the disease to prevent vision loss.

Early detection of DR through retina image analysis is an effective approach for identifying signs of the disease at its first stages. This approach involves analyzing fundus images to identify some characteristic signs of diabetic retinopathy, such as microaneurysms, hemorrhages, and exudates, which indicate the presence and severity of the disease.

- Microaneurysms (MAS) is an early sign of DR, resulting from the fragility of blood vessel walls. The retina can show small red circular dots measuring less than 125 µm and exhibit various sharp pointed margins [2], as illustrated in Figure 1.

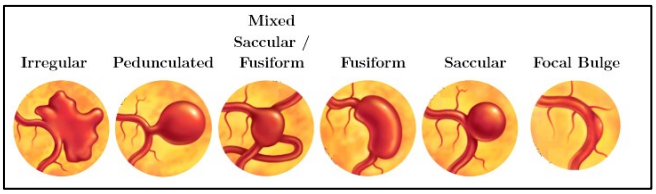

Figure 1: Microaneurysms (MA) types.

- Hemorrhage (HMS) shows up as larger irregular spots on the retina shown in Figure 2, with a size exceeding 125 µm [3].
- Hard exudates, depicted in Figure 2, manifested as yellow spots on the retina caused by plasma outflow, with sharp margins in the outer retina layers [4].
- Soft exudates, illustrated in Figure 2, resemble white spots resulting from nerve fiber swelling, typically showcasing an oval or round shape [2].

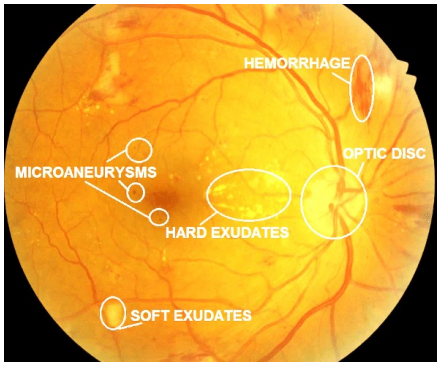

Figure 2: Illustration of various types of lesions in a retinal fundus image

Based on the Table 1, the stages of diabetic retinopathy (DR) are divided into five levels: No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR, the signs of each stage are also mentioned in the table below.

Table 1 :Stages and signs of the Diabetic Retinopathy

Stage	Severity Level	Dilated Ophthalmoscopy Observable Signs
1	No DR	No abnormalities
2	Mild DR	Microaneurysms only
3	Moderate DR	Micro-aneurysms, retinal dot and blot hemorrhages, hard exudates or cotton wool spots without signs of severe NPDR
4	Severe DR	More than 20 intraretinal hemorrhages in each of 4 quadrants Definite venous beading in two or more quadrants. Prominent IRMA in one or more quadrants
5	Proliferative DR	Neovascularization Vitreous/preretinal hemorrhage

IRMA = Intraretinal Microvascular Abnormalities.
NPDR = NonProliferative Diabetic Retinopathy.

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You can find a visual sampling of the different DR stages in Figure 3.

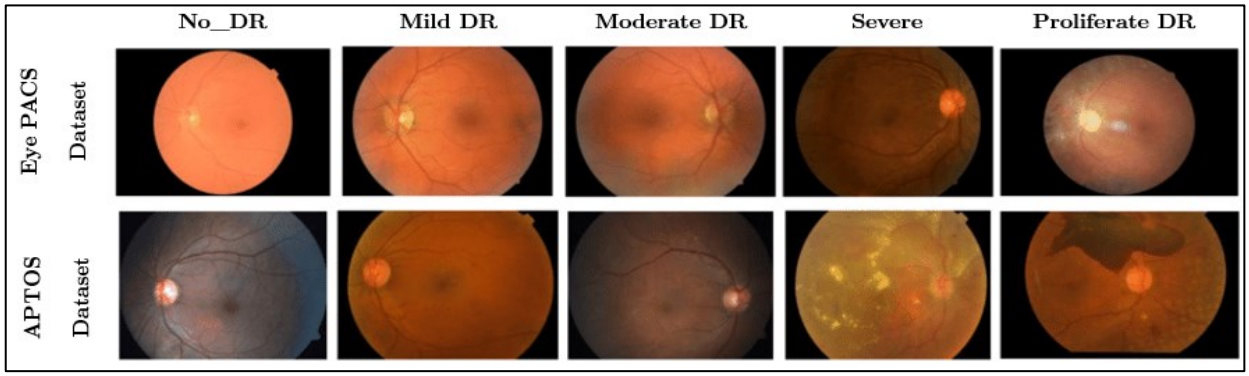

Figure 3: A sample of Different DR stages images.

It appears that is cheaper and quicker to use the automatic methods for DR detection than to do it manually. A manual diagnosis requires more effort than automated methods and is prone to misdiagnosis, since that human being is subject to error more than the machine. By utilizing advanced image analysis algorithms, healthcare professionals can accurately identify these signs and determine the appropriate course of treatment.

Artificial intelligence (AI) has a significant role in clinicians' decision-making, including diagnoses, treatment decisions and predicting results [5].

Deep neural networks, also known as brain-like deep learning systems, are able to automatically learn a large number of abstract high-level features directly from raw big data to ensure distributed representation of the data. Speech recognition and machine translation have been achieved with the Recurrent Neural Network (RNN), a widely used type of deep neural network, RNN has achieved also an unmatched success in academia and industries. Convolutional neural systems are profoundly specialized in sees for image recognition, analysis and classification, which makes them favored for the characteristics of the spatial coherence of this type of data, also DL's coverage of medical studies has grown in recent years. In addition, they have capacities better than those of medical experts, particularly in the classification for example of skin and breast cancers and the detection of lung cancer and Retinopathy of Prematurity (ROP).

However, there are still challenges when using Convolutional Neural Networks (CNNs) in medical research. First, it is difficult to obtain enough real medical images, notably for some specific diseases also the availability of labeled medical data is bounded. Tiny lesions of DR remain undetected in poor image quality because they are disposed to have cross-influences with other lesions. Labeling fundus photos is a manual and subjective process that is prone of errors, also limited medical image data and inevitable image noise make it difficult to achieve high accuracy in disease detection. Therefore, deep learning adopts two important strategies: transfer learning and ensemble learning, knowledge learned from one task can be used to enhance learning and performance on another related task is the basic concept of transfer learning, and the principle of the ensemble learning is to use a series of component algorithms with different learning preferences to solve the same prediction issue, those methods can achieve stronger generalization capabilities than a single algorithm.

In the current work, we have developed a model which takes advantage of the two techniques mentioned previously, namely transfer learning and ensemble stacking, the suggested model uses a pre-trained VGG-16 model for feature extraction and a stacking classifier model with nine algorithms for multi-class classification.

This article's organization is as follows: The second section will be dedicated to related works concerning detection and classification of DR disease, then in the third section, we will explain the methodology followed in our work, then in the fourth section, we will present the results of our experience, finally in the fifth section, we will end with a conclusion and some future perspectives.

II. RELATED WORKS

Over the past decade, the use of AI, including deep neural networks, has grown significantly in the field of DR detection. Previous works in this area has used deep learning architectures to classify fundus images based on the existence or absence of DR signs. CNNs are commonly employed in image recognition due to their capacity to capture intricate features and apply the findings to new images. In this section, we will review previous works using CNNs for DR detection and classification.

In a recent study conducted by Wei Zhang et al., an advanced system known as DeepDR was introduced for automated diabetic retinopathy (DR) identification and scoring. DeepDR utilizes transfer learning and ensemble learning techniques of DL models to directly identify the presence and stage of DR from retinal images. This system incorporates a collection of cutting-edge neural networks, including popular CNNs and custom standard DNNs, to achieve its objectives. The development of the DeepDR system involved the creation of a high-quality dataset of medical DR images, which were then meticulously labeled by clinical ophthalmologists [6].

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Trilok Nath Pandey et al. realized a study involving preprocessing and feature extraction from images, as well as training and evaluating deep learning models for categorization. The proposed model was able to deduce the RD phase from retinal fundus images, categorizing them into five stages. The overarching aim of the work was to develop a tool to aid healthcare professionals in diagnosis and treatment of DR, thereby improving patient outcomes. Two models were proposed in this study: a sequential model comprising thirteen CNN layers and a Resnet34 model with three CNN layers merged [7].

Cheena Mohanty et al. introduced two DL architectures for DR detection and classification: the idea is to design a hybrid network which combine VGG16 and XGBoost Classifier, and a DenseNet 121 network. The performance of these models was primarily evaluated by their accuracy. The study revealed that the hybrid network achieved an accuracy of 79.50% [8].

In a separate study by G. Kalyani et al., the detection and classification of diabetic retinopathy was done using a reformed capsule network. This model leveraged the convolution and primary capsule layer to extract features from retinal images, and the probability of an image belonging to a specific class was estimated using the class capsule layer and softmax layer. Furthermore, the proposed capsule network was exclusively trained with the Messidor dataset, which identifies DR in four stage grades from 0 to 3. Notably, the network was not trained on other datasets that classify diabetic retinopathy into five stages. The future direction of this research is to expand the training of the proposed capsule network to encompass all possible classes of DR [9].

In the in the study carried out by Wei Zhang and his team, an ensemble stacking model was used which included a set of deep learning models, and they trained their model over 300 epochs which need a lot of resources, to predict only four stages of DR, on the other hand in the second work Trilok Nath Pandey and his collaborators did not use a technique to balance its data which will leave the model effective only on the class whose number of images is greater, on its part Cheena Mohanty and his team opted for the boosting technique which reached a score of 79.5%, finally, G. Kalyani and his co-workers used only 1200 images belonging also to four distinct classes.

In our study, we bypassed all the shortcomings present in the studies cited above, first we worked on 3662 images belonging to five classes, moreover, our model requires less time in its execution given that it uses transfer learning and ensemble learning with a set of ML algorithms, and because we also used the Linear Discriminant Analysis (LDA) technique to reduce the size of the data and therefore reduce the execution time, and to avoid the problem of imbalanced data we used Synthetic Minority Oversampling Technique (SMOTE) technique which will allow us to make a fair classification and promote better precision for each class.

III. METHODOLOGY

Our approach steps shown in Figure 4 was to first use a VGG16 model to extract data from the preprocessed fundus images, then save them locally in a CSV file with the aim of not redoing the extraction therefore saving execution time, then use this data from this file and use techniques to balance and reduce the size of these extracted features before passing them to a stacking classifier model to make the prediction using the cross-validation method.

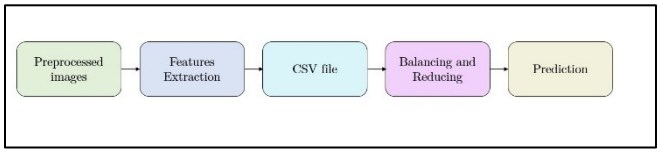

Figure 4: Steps of our Approach

A. The VGG-16model

The VGG-16 model, developed by the Visual Geometry Group at the University of Oxford, is a powerful CNN architecture with 16 layers. Its deep structure, along with a sequence of convolutional and max-pooling layers, enables it to excel in tasks like image classification and object recognition. This model's ability to learn complex visual representations has contributed to its widespread popularity and its success in the ImageNet Large Scale Visual Recognition Challenge where it achieved top ranks in 2014.

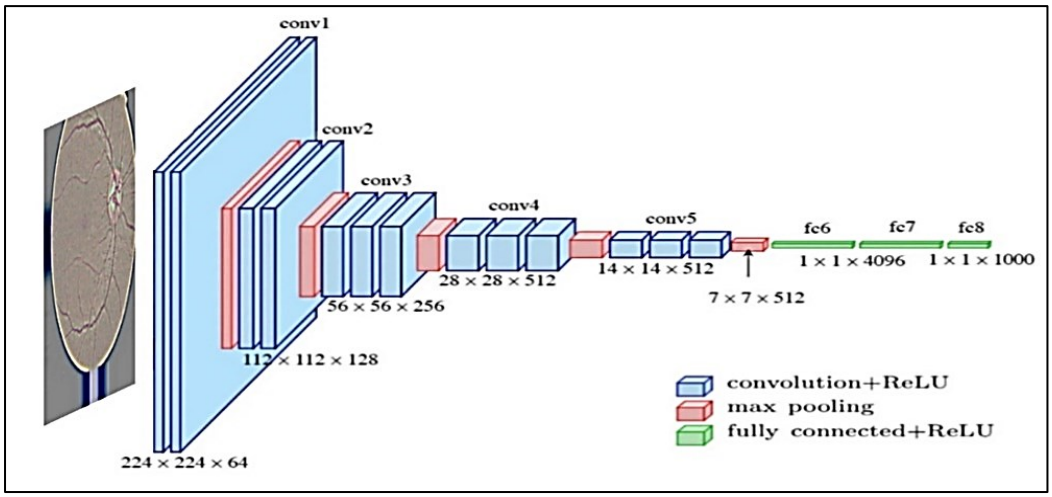

Figure 5: VGG-16 Architecture[10]

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VGG-16 has 16 layers that are weighted. In VGG-16, there are 13 convolutional layers, 5 Max Pooling layers, 3 Dense layers, which add up to 21 layers. However, it only has 16 weight layers, namely the learnable parameters layer. The default input size for the VGG-16 model is 224x224 with 3 RGB channels, all is shown on Figure 5 above.

B. Ensemble Learning – Stacking

Deep Neural Networks DNNs are employed in the classification tasks for supervised learning. The accuracy of these tasks is enhanced by other methods, such as ensemble learning methods, one of them is stacking [11].

The idea is "A group of people are collectively smarter than one individual expert". Combining the predictions from multiple individual models to get a better prediction performance is known as ensemble learning. Ensemble learning is a technique for aggregating the predictions of multiple assembled models, each of which may have its own strengths and weaknesses. This can result in enhanced efficiency and generalization. In this study, we chose the stacking technique for ensemble learning.

Stacking: An algorithm that takes predictions from base models and generates more precise predictions is used to combine the output of multiple base models. The combiner can be any machine learning algorithm, which allows for more flexibility in combining diverse models. In our case the meta classifier was Logistic Regression.

C. Dataset description

The data utilized in our study is centered around a dataset obtained from Kaggle [12], encompassing 3662 retinal scans.

In our study these retinal scans are used as images to identify DR. To make them compatible with a wide range of deep learning pre-trained models, these images have been scaled to 224 by 224 pixels. Every image has already been saved into the appropriate folder based on the degree or stage of DR.

The dataset categorizes retinal images into 5 classes reflecting different degrees of DR severity: Class 0 signifies non-DR, Class 1 embodies mild DR, Class 2 represents moderate DR, Class 3 indicates severe DR, and Class 4 denotes proliferative DR.

Table 2 outlines the number of images in each stage and shows how the various samples are distributed throughout these severity levels.

Table 2: Distribution of samples according to class and severity of DR.

Class	Severity	Number of Samples
0	Normal	1805
1	Mild	370
2	Moderate	999
3	Severe	193
4	Proliferative	295
Total		3662

D. Features Selection LDA

One of the well-known dimensionality reduction techniques in machine learning is Linear Discriminant Analysis (LDA), which is often used to address multi-class classification problems. LDA can change the size of the features from a higher dimensional space into a lower-dimensional space, helping to reduce resources and dimensional costs. In medical applications, LDA can assist in diagnosing diseases or conditions based on patient data. For example, it can be used to categorize patients as healthy or affected by a specific disease based on a set of medical data [13].

In our work, the size of our extracted features from the flatten layer was 25088, which is large and will require a lot of resources in terms of time and hardware for their processing, therefore we had to reduce this size with the LDA technique to speed up the processing, the Table 3 below shows the size before and after applying LDA.

Table 3: Size of Features before and after applying LDA.

	X 'Features'	Y 'Label'
Before LDA	(3662, 25088)	(3662,)
After LDA	(3662, 4)	(3662,)

E. SMOTE for imbalanced Class Handling

In our study, we addressed the issue of machine learning techniques displaying a bias towards the majority class while neglecting the minority class during classification, leading to misclassification of the minority class as the majority class. To rectify this disproportional distribution in the dataset, we applied the Synthetic Minority Oversampling Technique (SMOTE). SMOTE functions by generating new minority instances between existing ones by using linear interpolation, thereby creating virtual training records for the minority class. These synthetic training records are made by randomly selecting one or more of the k-nearest neighbors for each example in the minority class. The algorithm is used to sample the features in the minority class in the training dataset [14]. Following the oversampling process, various classification models were employed for the processed data.

As shown in Table 4, the dataset was unbalanced before using SMOTE. We were able to overcome this problem using the SMOTE technique, our dataset was balanced as shown in the table below.

Table 4: The count of images of each class before and after using SMOTE

Class	Count before SMOTE	Count after SMOTE
0	1805	1805
1	370	1805
2	999	1805
3	193	1805
4	295	1805
Total	3662	9025

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F. Model Architecture

The steps followed in designing our model are shown in Figure 6, firstly we bring in a number of **3362** fundus images, then preprocess those images, then pass this dataset to a pretrained VGG16 model using transfer learning for data extraction, after that we will pass this data to a stacking classifier to make prediction.

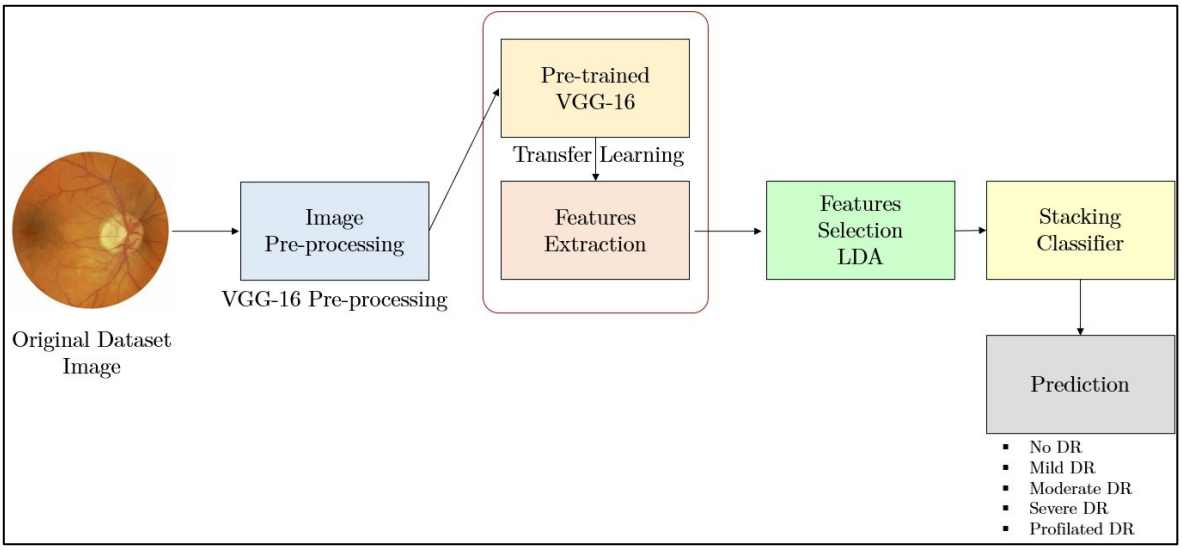

Figure 6: The conception of Our Model.

The architecture of our model is shown in Figure 7, the idea is to extract the data from the flatten layer in the form of a one-dimensional vector from the flatten layer in a CSV file. Given that the size of this vector is large '25088', as well as the data is not balanced, we used two techniques, first, a feature selection technique called LDA, to reduce the size of the extracted vector, second, another technique called SMOTE to balance the amount of data. All is shown previously in Table 3 and in Table 4.

After balancing and reducing features, we trained a stacking classifier with nine machine learning models using the cross-validation method for better results.

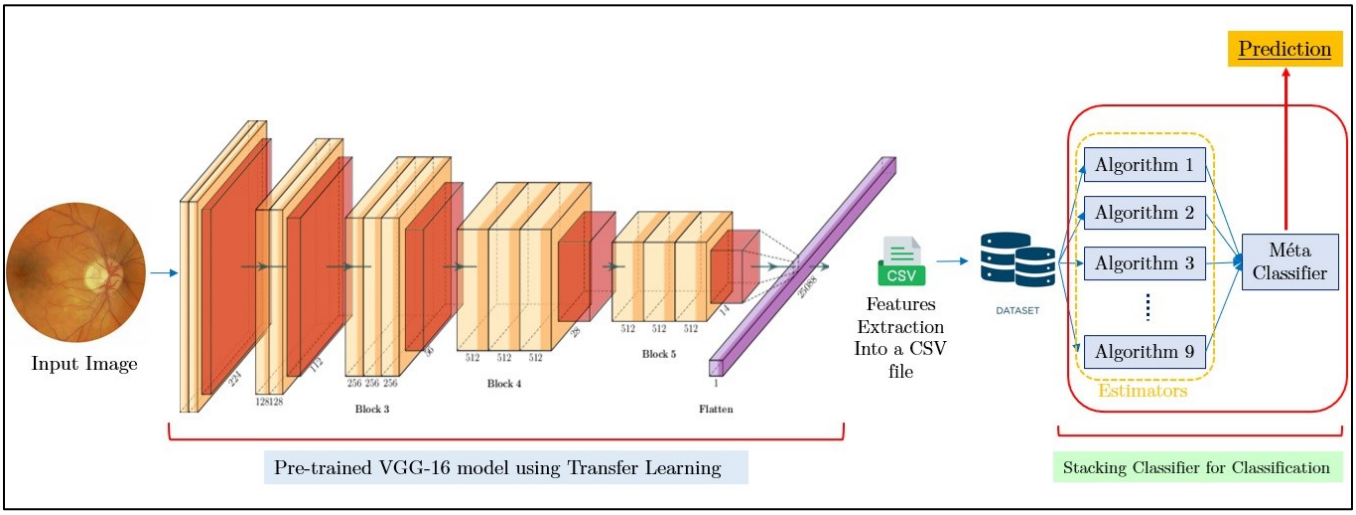

Figure 7: Architecture of the proposed Model.

IV. EXPERIMENTAL RESULTS

Below is an outline of the methodology in brief and the procedures we employed in the development and validation of our model:

1. We brought five classes fundus images then do the preprocessing.
2. We imported pre-trained weights from a network that has undergone training on the Imagenet dataset.
3. All weights in the lower convolutional layers were frozen.
4. Features were extracted from the flatten layer of the model.
5. A CSV file was used to save the extracted features.
6. The extracted data from the CSV file was then split into training and testing datasets.
7. We reduced the dimensionality of this dataset using the LDA technique.
8. An increase in the number of samples was achieved using the SMOTE technique to handle imbalanced dataset.

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9. To further enhance the classification process, a stacking method was implemented with nine machine learning algorithms.
10. Finally, the model was thoroughly fitted.

We applied the StratifiedKFold cross-validation approach to train the stacking classifier and validate the effectiveness of our model. Our model achieved an impressive 99.50% accuracy.

Compared to the studies cited in the third section, in ours we managed to fulfill the GAP existing in these works in terms of precision and number of images and their distribution with better accuracy, the Table 5 below show a brief comparison between the results and dataset description in studies cited in related works and ours.

Table 5: Comparison of the results and datasets between related works and our study

Paper	Dataset	Dataset Description	Classes	Accuracy
[6]	3833	imbalanced	4	97.15 %
[7]	3662	imbalanced	5	88.11 %
[8]	3662	balanced	5	79.50 %
[9]	1200	imbalanced	4	97.98 %
Our study	**3662**	**balanced**	**5**	**99.50 %**

The Figure 8 shows the resulting confusion matrix and Figure 9 expose the ROC curve illustrating the performance of our proposed model, other measurements that show that our solution can achieve high performance are shown in Figure 10.

As shown in Table 6, the stacking classifier technique turns out to be more efficient in terms of accuracy since it reached an average a bit greater than the precision if we use a single algorithm.

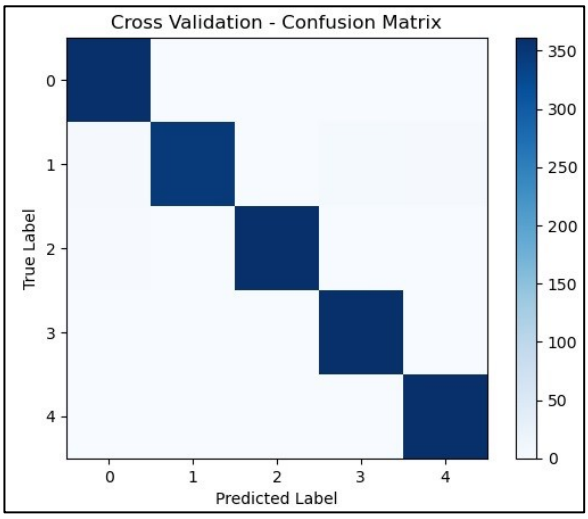

Figure 8: Confusion Matrix of our proposed model.

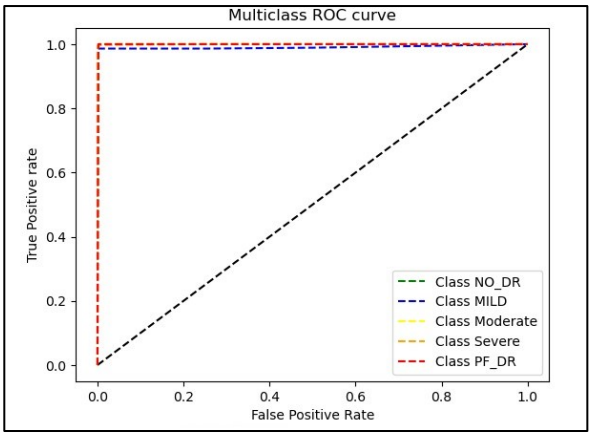

Figure 9: The ROC curve of our proposed model.

```
ROC AUC score - "one-vs-rest": [0.99859944 0.99126065 0.99659667 0.99860418 0.99897681]
ROC AUC score - "one-vs-one": 0.9968
```

Figure 10: AUC measurement

Table 6: Comparison between the accuracy of each single algorithm with the accuracy of stacking classifier

Algorithm	Accuracy
Random Forest	96.18 %
Gradient_boosting	93.57 %
SVM	93.14 %
Decision_tree	93.98 %
Extra_trees	96.86 %
Logistic_Regression	93.06 %
KNeighborsClassifier	95.51 %
GaussianNB	92.52 %
MLPClassifier	93.62 %
Stacking Classifier	**99.50 %**

V. CONCLUSION

In our research, we have proposed a novel multiclass classification approach for a fast and accurate detection of diabetic retinopathy disease utilizing a combination of Deep CNN transfer learning and ensemble stacking techniques with fundus images. A pre-trained VGG16 model was used to extract features, furthermore, we utilized an 'LDA' selection technique to reduce the dimensionality of these features and we solved the problem of imbalanced data by using SMOTE technique, for the classification phase, we employed a stacking method that combined nine machine learning algorithms and utilized cross-validation to maximize training data and enhance accuracy.

Our model's performance, evaluated on the Kaggle dataset, demonstrated a remarkable accuracy of 99.50%. In future work, we aim to explore other fundus image datasets of superior quality and other techniques and models in pursuit of achieving even better performance.

REFERENCES

[1] R. Taylor et D. Batey, *Handbook of Retinal Screening in Diabetes: Diagnosis and Management*, Wiley-Blackwell. 2012.

[2] M. Dubow *et al.*, « Classification of Human Retinal Microaneurysms Using Adaptive Optics Scanning Light

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Ophthalmoscope Fluorescein Angiography », *Investig. Opthalmology Vis. Sci.*, vol. 55, nº 3, p. 1299, 2014, doi: 10.1167/iovs.13-13122.

[3] N. Murugesan, T. Üstunkaya, et E. Feener, « Thrombosis and Hemorrhage in Diabetic Retinopathy: A Perspective from an Inflammatory Standpoint », *Semin. Thromb. Hemost.*, vol. 41, nº 06, p. 659-664, 2015, doi: 10.1055/s-0035-1556731.

[4] J. Amin, M. Sharif, et M. Yasmin, « A Review on Recent Developments for Detection of Diabetic Retinopathy », *Scientifica*, vol. 2016, p. 1-20, 2016, doi: 10.1155/2016/6838976.

[5] K. Elaissaoui et M. Ridouani, « Application of Deep Learning in Healthcare: A Survey on Brain Tumor Detection », *ITM Web Conf.*, vol. 52, p. 02005, 2023, doi: 10.1051/itmconf/20235202005.

[6] W. Zhang *et al.*, « Automated identification and grading system of diabetic retinopathy using deep neural networks », *Knowl.-Based Syst.*, vol. 175, p. 12-25, 2019, doi: 10.1016/j.knosys.2019.03.016.

[7] T. N. Pandey, S. S. Pattnaik, S. Sharma, V. Ravalekar, B. B. Dash, et S. S. Patra, « An Empirical Study of Deep Learning Techniques for Diabetic Retinopathy Detection », in *2023 3rd International Conference on Innovative Sustainable Computational Technologies (CISCT)*, Dehradun, India: IEEE, 2023, p. 1-6. doi: 10.1109/CISCT57197.2023.10351463.

[8] C. Mohanty *et al.*, « Using Deep Learning Architectures for Detection and Classification of Diabetic Retinopathy », *Sensors*, vol. 23, nº 12, p. 5726, 2023, doi: 10.3390/s23125726.

[9] G. Kalyani, B. Janakiramaiah, A. Karuna, et L. V. N. Prasad, « Diabetic retinopathy detection and classification using capsule networks », *Complex Intell. Syst.*, vol. 9, nº 3, p. 2651-2664, 2023, doi: 10.1007/s40747-021-00318-9.

[10] K. Simonyan et A. Zisserman, « Very Deep Convolutional Networks for Large-Scale Image Recognition ». arXiv, 2015. doi: 10.48550/arXiv.1409.1556.

[11] A. Elaanba, M. Ridouani, et L. Hassouni, « Automatic Diagnosis Chest-X-Ray-Based-Framework for Semantic Segmentation fnd Placement Errors Detection of Catheters and Tubes », 2023.

[12] « Diabetic Retinopathy 224x224 (2019 Data) ». Disponible sur: https://www.kaggle.com/datasets/sovitrath/diabetic-retinopathy-224x224-2019-data

[13] C. Techa, M. Ridouani, L. Hassouni, et H. Anoun, « Automated Alzheimer's Disease Classification from Brain MRI Scans Using ConvNeXt and Ensemble of Machine Learning Classifiers », in *Proceedings of the 14th International Conference on Soft Computing and Pattern Recognition (SoCPaR 2022)*, A. Abraham, T. Hanne, N. Gandhi, P. Manghirmalani Mishra, A. Bajaj, et P. Siarry, Éd., Cham: Springer Nature Switzerland, 2023, p. 382-391. doi: 10.1007/978-3-031-27524-1_36.

[14] J. Rabbah, M. Ridouani, et L. Hassouni, « A New Classification Model Based on Transfer Learning of DCNN and Stacknet for Fast Classification of Pneumonia Through X-Ray Images »:, *Int. J. Reliab. Qual. E-Healthc.*, vol. 12, nº 1, p. 1-23, 2023, doi: 10.4018/IJRQEH.326765.

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